

- Introducing one air bubble is not very easy.
- The twig may not remain alive for very long.

- a) doesn't move much
- b) moves toward plant
- c) moves toward plant

(iii) White light as photosynthesis will be maximum.

(b) New starch formed will be more prominent Shows production of Starch by photosynthesis.

(iv) Stroma - Dark reaction  
Granum - light reaction  
Stroma - colourless ground substance  
Granum - stack of thylakoids.

(v) Activation of chlorophyll, splitting of water.

2.2  
(i) Potometer (Jannong's)

(ii) Used to measure uptake of water by the shoot of a plant. The air bubble in the potometer moves with the uptake of water by the plant and the distance travelled by the bubble in a given time is the rate of transpiration.

(iii)  
(a) The reservoir plays an important role as it is useful for setting the air bubble to start position.

(b) Fill the entire apparatus with water.

(iii) Law of independent assortment: When there are two pairs of characters, the distribution of the alleles of one character into the gametes is independent of the distribution of the alleles of the other character.

New types of plants with new combinations of useful characters can be produced by hybridisation.

(iv) Respiration is said to be the reverse of photosynthesis because in respiration, oxygen helps break down sugars to obtain energy and carbon dioxide is released, while in photosynthesis carbon dioxide and water in presence of sunlight and chlorophyll forms ATP and oxygen.



(v) (a)

(b)

(iv)

(v)

22

(i)

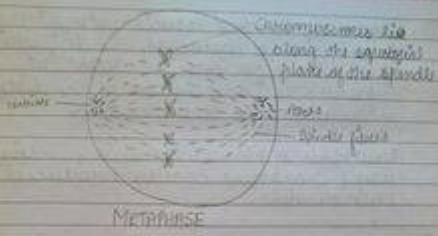
(ii)

(iii)

(a)

(b)

Q.1



METAPHASE

Q.2

- (a) Mitosis  
(b) Meiosis

| (i) | Mitosis                                                | Meiosis                                                     |
|-----|--------------------------------------------------------|-------------------------------------------------------------|
| →   | Single nuclear division                                | Two nuclear divisions                                       |
| →   | Two daughter cells                                     | Four daughter cells                                         |
| →   | Full set of chromosomes are passed onto daughter cells | Half the number of chromosomes are passed to daughter cells |

Q.3

(i) 9:3:3:1

|    |      |      |      |      |
|----|------|------|------|------|
|    | RY   | Ry   | rY   | ry   |
| RY | RRYY | RRYy | RrYY | RrYy |
| Ry | RRYy | RRyy | RrYy | Rryy |
| rY | RrYY | RrYy | rrYY | rrYy |
| ry | RrYy | Rryy | rrYy | rryy |

(c) Crossing over is the exchange of genes between homologous chromosomes, resulting in a mixture of parental features.

(d) Karyokinesis is the division of nucleus during mitosis.

(e) Interphase is the resting phase between successive mitotic divisions of a cell, or between the first and second divisions of meiosis.

(v) The adaptation of leaves for photosynthesis are:

- leaves provide large surface area for maximum light absorption.
- leaves are arranged at right angles to the light source in a way that causes overlapping.
- the extensive network of veins enables quick transport of substances to and from the mesophyll cells.
- Presence of numerous stomata for gaseous exchange.
- The chloroplasts are more in number on the upper surface of leaves.

### Section B

18. (i) Prophase

(ii) 1 - centriomere, 2 - chromosome, 3 - spindle fibres, centriole form arcs.

Each single DNA strand has repeating nucleotides. They are made up of sugar, phosphate and a nitrogenous base attached to the sugar which extends to join the nitrogenous base from the other strand. The two strands together make a ladder-like arrangement with the nitrogenous bases forming 'rungs' of the ladder.

(10)

(a) Detached plant is the plant which lacks starch. Autotrophs are done by keeping the plant in dark for 2-3 days for converting a part of a leaf with a black paper by clip.

(b) The leaf is plucked and is boiled in water and decolourized with alcohol.

→ The portion of leaf which received light turns blue with iodine because photosynthesis took place in that part of leaf.

→ The part of leaf which was covered with black paper remains colourless because photosynthesis could not take place in absence of sunlight.

(11)

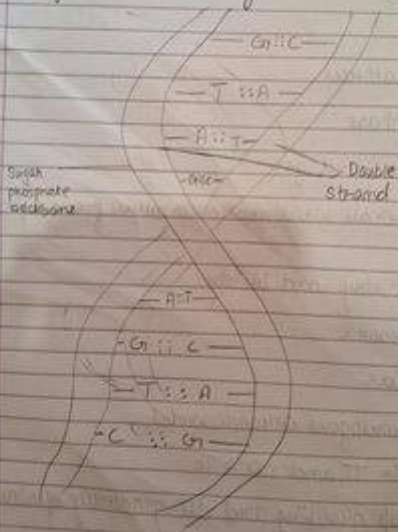
(a) Mitosis is the cell division leading to the production of diploid cells for growth and development.

(b) Meiosis is the cell division leading to the production of haploid cells for growth and development.

16

- (1) Law of Dominance :- Out of a pair of contrasting characters present together, only one is able to express itself while the other remains suppressed. An example is a person's dominant trait which is being tall expressed itself more than being short which is recessive.

(ii)



# Biology

Section: A

- 1 (2) Peltic
- 2 (3) Sucrose
- 3 (1) Haploid set of chromosomes
- 4 (1)  $\text{CO}_2$
- 5 (3)  $\text{K}^+$
- 6 (1) Hypostomes
- 7 (1) Prophase
- 8 (3)  $\text{O}_2$  and  $\text{g}$ s
- 9 (3) Overall water molecules move from left to right
- 10 (1) Hot, dry and windy
- 11 (3) Protona
- 12 (2) Asters
- 13 (3) Homologous chromosomes
- 14 (3) 50% TF and 50% TE
- 15 (1) Stop growing and die gradually of starvation